

Immune Genes Used for Prediction		
Gene	Function	Category
CXCL9	T cell recruitment, IFN- γ -inducible	Chemokine
TNF	Pro-inflammatory, induces cachexia	Cytokine
IDO1	Tryptophan depletion, immunosuppression	Metabolic enzyme
IFN γ	Th1 master regulator, parasite control	Cytokine
IL-6	Acute phase response, inflammation	Cytokine
TICAM1	TLR adaptor, viral response	Innate signaling
RETNLB	Th2 effector, goblet cell hyperplasia	Th2 response
IRGM1	Autophagy, parasite resistance	Cell-autonomous
SOCS1	Negative feedback, prevents tissue damage	Regulatory
MUC2	Intestinal barrier, pathogen exclusion	Barrier
MPO	Oxidative burst, neutrophil marker	Innate effector
IL1RN	IL-1 antagonist, limits inflammation	Regulatory
IL-10	Anti-inflammatory, regulatory	Regulatory
IL-13	Th2 cytokine, mucus production	Th2 response
NCR1	NK cell activation receptor	Cytotoxic
PRF1	Cytolytic granule protein	Cytotoxic
TLR2	Bacterial recognition	Pattern recognition
TLR4	LPS sensor, innate immunity	Pattern recognition
ARG1	M2 macrophage marker, tissue repair	Metabolic enzyme